

MINISTRY OF EDUCATION AND RESEARCH
OF THE REPUBLIC OF MOLDOVA

Technical University of Moldova
Faculty of Computers, Informatics and Microelectronics
Department of Software Engineering and Automatics

Mihail Pancenco FAF-243

Report

Laboratory Work No. 2

Cryptography

Cryptanalysis of monoalphabetic ciphers

Frequency analysis

Variant 19

Supervisors:

Poștaru Andrei, *Doctor*

Maia Zaica, *University assistant*

DISA, FCIM, UTM

Chișinău – 2026

Table of Contents

1 Task and short theory **3**

1.1 Laboratory task and selected variant 3

1.2 Monoalphabetic substitution and frequency analysis 3

2 Initial frequency analysis **4**

2.1 Letter counts and comparison with English 4

2.2 Frequency distributions and letter combinations 5

3 Step-by-step recovery of the substitution **6**

3.1 The first seven substitutions from my working notes 6

3.2 Recognising KEY, FROM, FOR, IN and AND 7

3.3 Recovering SHOULD and BE 7

3.4 Completing the technical vocabulary and rare letters 8

3.5 Completing the alphabet and checking consistency 8

4 Recovered key and results **9**

4.1 The substitution alphabets 9

4.2 Verification and interpretation 10

4.3 Complete recovered plaintext 11

5 Conclusions **13**

A Complete working ciphertext **14**

Bibliography **15**

1 Task and short theory

1.1 Laboratory task and selected variant

The objective of Laboratory Work No. 2 is to recover an English plaintext from a monoalphabetic substitution cipher using letter frequencies, common letter combinations and the structure of words. My number in the group list is 19, so I analysed variant V19. The assignment specifies that only letters are encrypted; digits, punctuation and other characters remain unchanged. The report follows the worked attack in Section 2.3 of the assignment [1]. A program implementation is not required.

The supplied ciphertext begins with

```
CINZ WQVPV WRN CDGOTZVGWTS UIXGHXUSVP CNI PVSVHWXGJ DPTASV CXVSOHXUQVIP,
LVIHLQNCCP OVODHVO PXY PUVHXCXH IVBDXIVZVGWP
```

Its 3,230 letters match the letters of variant V19 in the assignment. The complete working ciphertext is included in Appendix A.

1.2 Monoalphabetic substitution and frequency analysis

A monoalphabetic substitution uses one fixed permutation π of the 26-letter English alphabet. Each plaintext letter m_i is replaced by $c_i = \pi(m_i)$, and decryption applies $m_i = \pi^{-1}(c_i)$. There are $26! \approx 4.03 \times 10^{26}$ possible alphabet permutations, but a large key space alone does not make this cipher secure.

The substitution changes letter names while preserving their counts, repeated-letter patterns and positions. If a cipher symbol occurs n_a times in a message containing N letters, its relative frequency is

$$f(a) = \frac{n_a}{N} \times 100\%.$$

I compared the ciphertext frequencies with the English reference values in Table 2.2 of the assignment. Frequent English letters include E, T, A, O, I, N and S. These are useful hypotheses, rather than a rule that the first-ranked ciphertext symbol must always represent E.

I also examined digraphs such as TH and HE, trigraphs such as THE and AND, double letters, and short words. A consistent substitution must give plausible words throughout the message and must not assign two distinct cipher symbols to the same plaintext letter. Confirmed plaintext letters are shown in lowercase in the intermediate states; undecoded cipher symbols remain uppercase.

2 Initial frequency analysis

2.1 Letter counts and comparison with English

I entered the variant into the frequency-analysis service recommended in the assignment, Crypto Corner [2], and used *Find Frequencies*. The results agree with the frequency values in my working notes. There are $N = 3230$ letters and 25 distinct cipher symbols; M does not occur. Table 2.1 gives exact counts and percentages rounded to two decimal places. Spaces, line breaks, digits and punctuation are excluded from this denominator.

Table 2.1 - Ciphertext frequencies in descending order

Letter	Count	%	Letter	Count	%
V	418	12.94	U	95	2.94
W	322	9.97	D	88	2.72
P	247	7.65	F	88	2.72
T	229	7.09	C	72	2.23
X	221	6.84	A	64	1.98
N	215	6.66	J	49	1.52
G	197	6.10	L	39	1.21
I	192	5.94	R	31	0.96
Q	167	5.17	K	20	0.62
S	143	4.43	B	11	0.34
H	118	3.65	Y	9	0.28
O	97	3.00	E	1	0.03
Z	97	3.00	M	0	0.00

For example, V occurs 418 times, so $f(V) = 418/3230 \times 100 = 12.94\%$, close to the English reference frequency of E, 12.70%. W occurs 322 times, or 9.97%, making T a plausible match. P occurs 247 times, or 7.65%. Its high rank alone might suggest A or O, but word patterns later identify it as S. This illustrates why replacing letters solely by frequency rank would be unreliable.

2.2 Frequency distributions and letter combinations

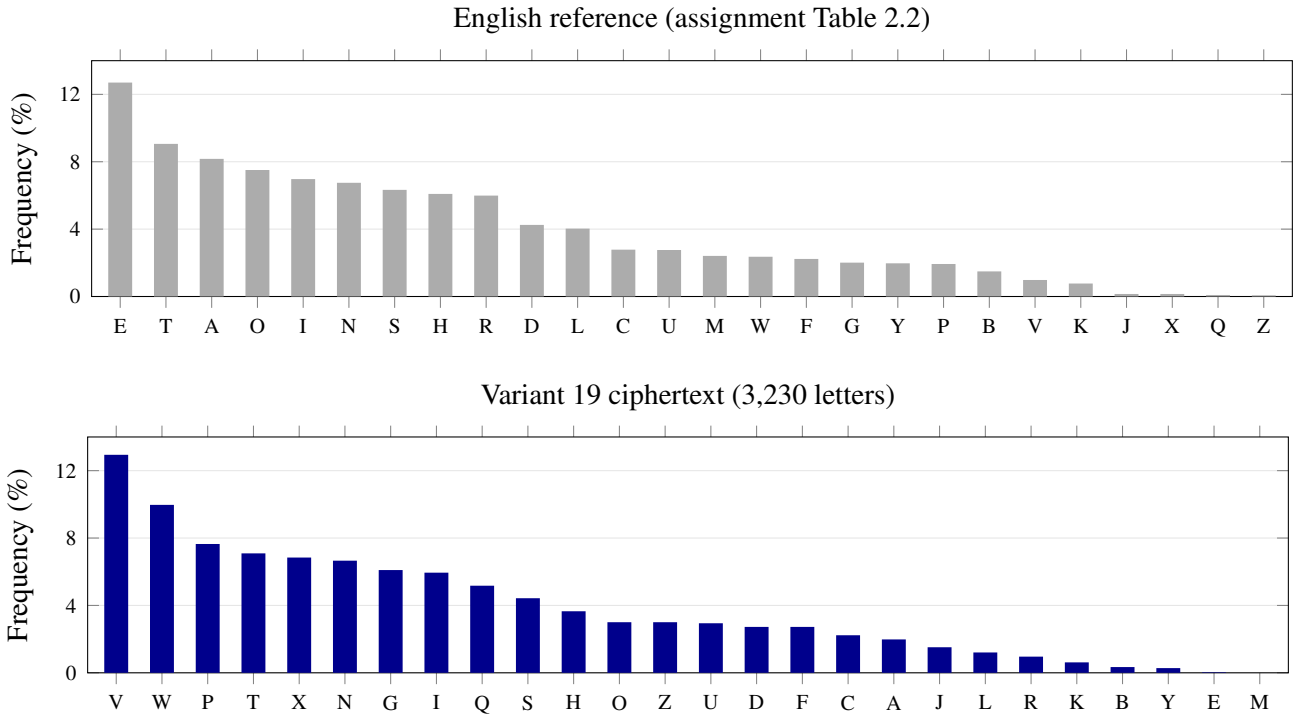


Figure 2.1 - English reference and variant 19 frequencies, each independently sorted

The two graphs show a similar overall frequency profile. Their horizontal positions represent frequency rank, so vertically aligned bars are not necessarily the same plaintext letter. Word and n-gram evidence is needed to establish the actual correspondence.

The service's *Count Digraphs*, *Count Trigraphs* and *Count Doubles* options produced the combinations below. For the counts shown in this report, combinations are counted within uninterrupted runs of letters, without joining across spaces or punctuation. Missing spaces already present in the copied variant are left intact.

Table 2.2 - Most frequent letter combinations in variant 19

Type	Ciphertext combinations, in the service's order
Digraphs	`WQ, QV, VI, XG, IV, PW, NG, SV, XW, TG, QT, VP, VG`
Trigraphs	`WQV, QTW, WQT, ASV, VGW, XGJ, QVI, TGO, WXN, XNG, FPW, NDS, DSO`
Doubles	`SS, PP, WW, CC, VV, II, UU, NN, ZZ, HH, GG`

3 Step-by-step recovery of the substitution

3.1 The first seven substitutions from my working notes

I started with the frequent trigram WQV, which occurs 63 times inside letter runs and appears 44 times as a separate word. The English word THE is a strong candidate. This also fits the frequent digraphs WQ (91 occurrences) and QV (81 occurrences).

My recorded first substitution [3] was W→T. I then added Q→H and V→E, one at a time, obtaining the from WQV. The opening after these three substitutions is

```
CINZ thePe tRN CDGOTZeGtTS UIXGHXUSeP CNI PeSeHtXGJ DPTASe CXeSOHXUheIP, LeIHLhNCCP OeODHeO
PXY PUeHXCXH IeBDXIeZeGtP: (1) the PFPteZPhNDSO Ae, XC GNt theNIetXHTSSF DGAIeTLTASe,
DGAIeTLTASe XG UITHtXHe; (2)HNZUINZXPe NC the PFPteZ PhNDSO GNt XGHNGKeGXeGHe the
```

Next, WQVPV became thePe, which suggested THESE. I replaced P with S, obtaining these. This was checked against PFPWVZ, which became sFsteZ, a likely instance of SYSTEM. The next substitution, F→Y, gave systeZ; it also changed LVF into LeY, suggesting KEY. Replacing Z with M completed system. Finally, I replaced T with A: WQTW became that, and other partial words gained the expected A positions.

Table 3.1 - Recorded order of the first seven substitutions

Step	Substitution	Main evidence
1	W→T	WQV is a candidate for THE
2	Q→H	tQV becomes thV
3	V→E	thV becomes the; V is the most frequent symbol
4	P→S	thePe becomes these
5	F→Y	sFsteZ becomes systeZ; LeF becomes Ley
6	Z→M	systeZ becomes system
7	T→A	thTt becomes that

After these seven substitutions, the opening is

```
CINm these tRN CDGOameGtaS UIXGHXUSes CNI seSeHtXGJ DsaASe CXeSOHXUheIs, LeIHLhNCCs OeODHeO
sXY sUeHXCXH IeBDXIemeGts: (1) the systemshNDSO Ae, XC GNt theNIetXHaSSy DGAIeaLaASe,
DGAIeaLaASe XG UIaHtXHe; (2)HNmUINmXse NC the system shNDSO GNt XGHNGKeGXeGHe the
```

The supplied working notes stop at T→A. The following sections complete the analysis using the same word-pattern method; they present a reproducible continuation rather than additional recorded dialogue.

3.2 Recognising KEY, FROM, FOR, IN and AND

The word LVF now reads Ley. In the repeated phrase the Ley, KEY is the natural hypothesis, so I add L→K. This is later checked in the longer phrase *the specific key*.

The opening CINm these tRN suggests *from these two*. FROM gives C→F, I→R and N→O. The separate word CNI then becomes for, and NC becomes of, supporting all three assignments in several contexts.

```
from these tRo fDGOameGtaS UrXGHXUSes for seSeHtXGJ DsaASe fXeSOHXUhers, kerHkhoffs OeODHeO
sXY sUeHXfXH reBDXremeGts: (1) the systemshoDSO Ae, Xf Got theoretXHaSSy DGAreakaASe,
DGAreakaASe XG UraHtXHe; (2)HomUromXse of the system shoDSO Got XGHoGKeGXeGHe the
```

The short word XW now reads Xt. IT is a plausible match, so I add X→I. The repeated pair XG becomes iG, suggesting IN and therefore G→N. This also makes GNW read not. The frequent trigram TGO now reads anO; AND supplies O→D.

These hypotheses are checked together, rather than accepted from one isolated word: XG→IN, XW→IT, GNW→NOT and TGO→AND all produce sensible English in the surrounding sentences.

3.3 Recovering SHOULD and BE

With the previous substitutions, PQNDSO reads shoDSd. The repeated construction the system shoDSd Ae strongly suggests *the system should be*. SHOULD gives D→U and S→L; BE gives A→B. The double SS becomes LL, which also fits TSS→ALL and the ending of FULFILLED.

At this point 17 cipher symbols have been recovered. The opening becomes

```
from these tRo fundamental UrinHiUles for seleHtinJ usable fieldHiUhers, kerHkhoffs deduHed
siY sUeHifiH reBuirements: (1) the systemshould be, if not theoretiHally unbreakable,
unbreakable in UraHtiHe; (2)HomUromise of the system should not inHonKenienHe the
```

The emerging words fundamental, principles and requirements provide longer checks on the same assignments.

3.4 Completing the technical vocabulary and rare letters

The text repeatedly refers to ciphers and their keys. In that context, HIFUWNJITUQVI is a candidate for CRYPTOGRAPHER. Its known letters already fit, and it supplies H→C, U→P and J→G. The same assignments are checked in HXUQVI→CIPHER, UITHWXHV→PRACTICE and HIFUWNJITZP→CRYPTOGRAMS. The double CC becomes FF, supporting the repeated name KERCKHOFFS after the rest of that name is completed.

The pattern RQXHQ now reads Rhich. WHICH gives R→W. This also completes WRN→TWO at the beginning and RXWQNDW→WITHOUT. The partial word QTKV reads haKe; HAVE supplies K→V.

```
from these two fundamental principles for selecting usable fieldciphers, kerckhoffs deduced
siY specific reBuirements: (1) the systemshould be, if not theoretically unbreakable,
unbreakable in practice; (2)compromise of the system should not inconvenience the
```

Three observed symbols remain. VYUSXHXW reads eYplicit, which suggests EXPLICIT and gives Y→X. The word WVHQGXBDV reads techniBue, so TECHNIQUE gives B→Q. Finally, the phrase containing WNHNGEDIV reads toconEure; restoring the word boundary gives *to conjure*, so E→J. E occurs only once, which explains why frequency ranking alone was insufficient to recover J.

```
from these two fundamental principles for selecting usable fieldciphers, kerckhoffs deduced
six specific requirements: (1) the systemshould be, if not theoretically unbreakable,
unbreakable in practice; (2)compromise of the system should not inconvenience the
```

The entire passage is now readable. These final substitutions are checked in the full message, including EXCEPTIONS, REQUIREMENTS and CONJURE. A readable result everywhere is stronger evidence than a plausible replacement in just one short word.

3.5 Completing the alphabet and checking consistency

All 25 symbols that occur in the message now have a consistent plaintext letter. The cipher symbol M has count zero. Under the assumption that the cipher uses a permutation of all 26 English letters, the only unused plaintext letter is Z; hence M→Z follows by elimination. This pair is inferred from alphabet completion and is not directly observed in V19. It has no effect on decoding this message.

The frequent patterns now have the following meanings: WQ→TH, QV→HE, WQV→THE, QTW→HAT, WQT→THA, TGO→AND, ASV→BLE and XGJ→ING. The doubles become LL, SS, TT, FF, EE, RR, PP, OO, MM, CC and NN. These are consistent with the words in the recovered English passage.

4 Recovered key and results

4.1 The substitution alphabets

Table 4.1 lists the decryption key: the upper row is the cipher alphabet and the lower row is the corresponding plaintext alphabet. All 26 output letters are distinct.

Table 4.1 - Recovered decryption mapping, cipher letter to plaintext letter

Cipher	A	B	C	D	E	F	G	H	I	J	K	L	M
Plain	B	Q	F	U	J	Y	N	C	R	G	V	K	Z*
Cipher	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Plain	O	D	S	H	W	L	A	P	E	T	I	X	M

*M→Z is inferred by elimination because M does not occur.

For re-encryption, the inverse mapping is

```
Plain:  ABCDEFGHIJKLMNOPQRSTUVWXYZ
Cipher: TAHOVCJQXELSZGNUBIPWDKRYFM
```

The key was also entered into the online service and applied with *Make Substitutions*. The site's complete output agrees with the independently checked decryption. Figure 4.1 shows the opening of the recovered message and the filled substitution row.

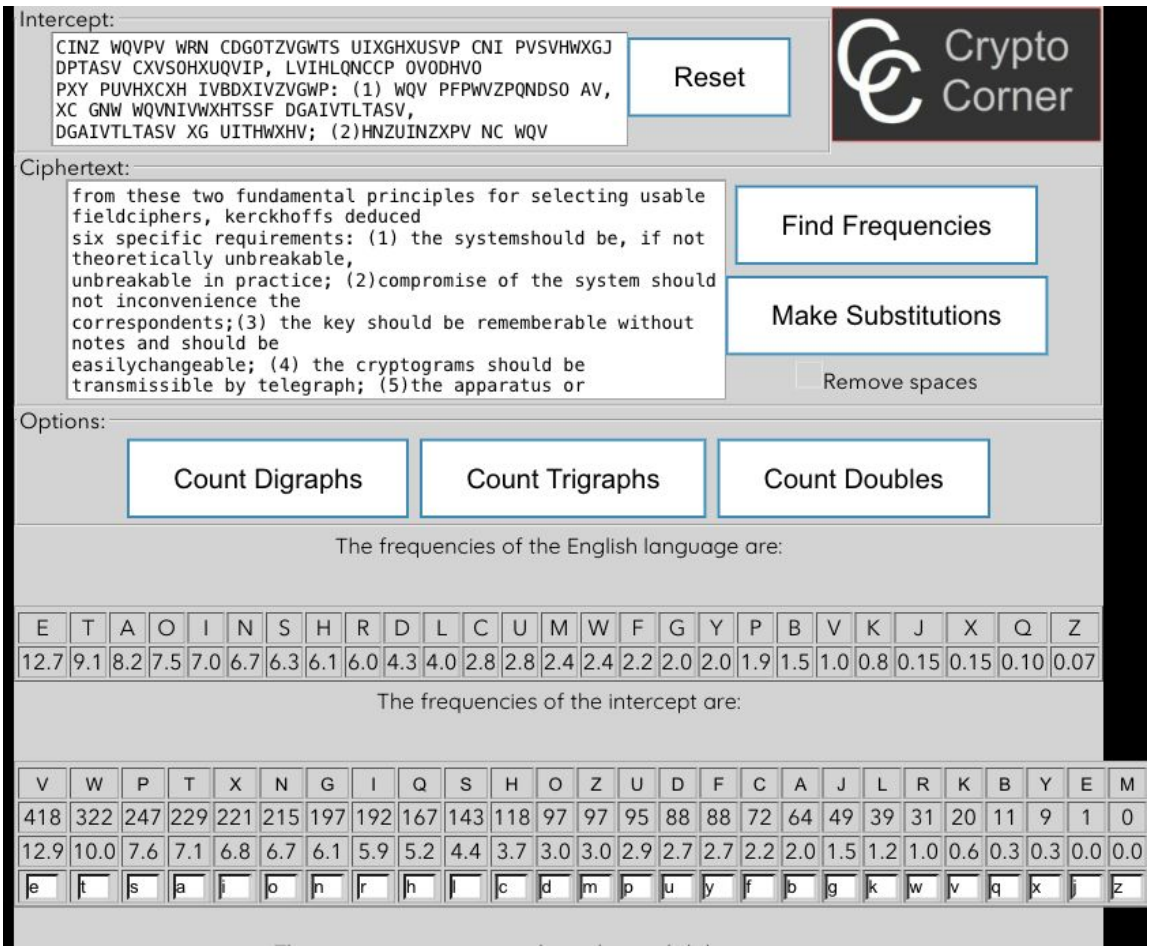


Figure 4.1 - Variant 19 decoded in the frequency-analysis service

4.2 Verification and interpretation

The mapping is one-to-one, all 25 observed symbols are decoded, and the English text is coherent. Applying the inverse key reproduces the working ciphertext exactly, including punctuation, digits, spaces and line breaks. The original supplied paste is also reproduced exactly when its extraction artifacts are retained.

The passage discusses Kerckhoffs' six requirements, the distinction between a public system and a secret key, and superimposition of polyalphabetic messages. These are topics in the recovered text; the attacked cipher is monoalphabetic.

The PDF paste has merged words, a control separator in POST-TELEGRAPH and a trailing backslash. The working input displays the separator as a hyphen and excludes that backslash; the supplied version is retained separately. The readable plaintext restores spacing, paragraphs and capitalization, and omits the stray hyphen in *have -been made*. All 3,230 letters remain identical to the exact decryption. Exact forms are saved in `data/plaintext_exact.txt` and `data/plaintext_supplied_exact.txt`.

4.3 Complete recovered plaintext

From these two fundamental principles for selecting usable field ciphers, Kerckhoffs deduced six specific requirements: (1) the system should be, if not theoretically unbreakable, unbreakable in practice; (2) compromise of the system should not inconvenience the correspondents; (3) the key should be rememberable without notes and should be easily changeable; (4) the cryptograms should be transmissible by telegraph; (5) the apparatus or documents should be portable and operable by a single person; (6) the system should be easy, neither requiring knowledge of a long list of rules nor involving mental strain.

These requirements still comprise the ideal which military ciphers aim at. They have been rephrased, and qualities that lie implicit have been made explicit. But any modern cryptographer would be very happy if any cipher fulfilled all six. Of course, it has never been possible to do that. There appears to be a certain incompatibility among them that makes it impossible to institute all of them at once. The requirement that is usually sacrificed is the first.

Kerckhoffs argued strongly against the notion of a field cipher that would simply resist solution long enough for the orders it transmitted to be carried out. This was not enough, he said, declaring that "the secret matter in communications sent over a distance very often retains its importance beyond the day on which it was transmitted." He was on the side of the angels, but a practical field cipher that is unbreakable was not possible in his day, nor is it today, and so military cryptography has settled for field ciphers that delay but do not defeat cryptanalysis.

Perhaps the most startling requirement, at first glance, was the second. Kerckhoffs explained that by "system" he meant "the material part of the system; tableaux, code books, or whatever mechanical apparatus may be necessary," and not "the key proper." Kerckhoffs here makes for the first time the distinction, now basic to cryptology, between the general system and the specific key. Why must the general system "not require secrecy," as, for example, a codebook requires it? Why must it be "a process that. . . our neighbors can even copy and adopt"?

Because, Kerckhoffs said, "it is not necessary to conjure up imaginary phantoms and to suspect the incorruptibility of employees or subalterns to understand that, if a system requiring secrecy were in the hands of too large a number of individuals, it could be compromised at each engagement in which one or another of them took part." This has proved to be true, and Kerckhoffs' second requirement has become widely accepted under a form that is sometimes called the fundamental assumption of military cryptography: that the enemy knows the general system. But he must still be unable to solve messages in it without knowing the specific key. In its modern formulation, the Kerckhoffs doctrine states that secrecy must reside solely in the keys.

Had Kerckhoffs merely published his perceptions of the problems facing post-telegraph cryptography and his prescriptions for resolving them, he would have assured a place for himself in the pantheon of cryptology. But he did more. He contributed a technique of cryptanalysis that is of supreme importance today. Called "superimposition," it constitutes the most general solution for polyalphabetic substitution systems. With few exceptions, it lays no restrictions on the type or length of keys, as does the Kasiski method, nor on the alphabets, which may be interrelated or entirely independent. It wants only several messages in the same key.

The cryptanalyst must align these one above the other so that letters enciphered with the same keyletter will fall into a single column. In the simplest case, that of a running key (a very long continuous text used as a key, as a novel) that restarts with each message, he can do this simply by placing all the first letters in the first column, all the second letters in the next column, and so on.

5 Conclusions

I recovered the plaintext of variant 19 through frequency analysis and successive letter substitutions. The first seven recorded replacements established THE, THESE, SYSTEM and THAT. Continuing with short words, double letters and the emerging technical vocabulary allowed the remaining observed letters to be identified. The full substitution was consistent throughout the passage and reproduced the ciphertext when reversed.

The work demonstrates that letter frequencies are useful starting points, but they do not determine the key on their own. For example, the third-most frequent cipher symbol P represents S, while the rare symbol E represents J. Digraphs, trigraphs, word patterns and sentence meaning supplied the additional evidence needed to confirm these mappings.

I also learned to distinguish an observed substitution from one inferred only by elimination: M does not appear in the ciphertext, so M$\rightarrow$Z completes the assumed alphabet permutation without being directly tested by the message. Preserving the exact input and decrypted versions makes it possible to separate cryptographic correctness from corrections to copied spacing and formatting.

Although monoalphabetic substitution has a large theoretical key space, it preserves enough language structure to be vulnerable to this attack. The main skills developed in this laboratory work were comparing frequency distributions, testing hypotheses across several words, maintaining a consistent substitution table and verifying a recovered plaintext. No laboratory program was required.

A Complete working ciphertext

The following is the full V19 input used for the frequency analysis. Capitalization and the copied word boundaries are preserved. The extraction separator is displayed as a hyphen and the trailing paste backslash is excluded, as explained in the verification section.

```
CINZ WQVPV WRN CDGOTZVGWTS UIXGHXUSVP CNI PVSVHWXGJ DPTASV CXVSOHXUQVIP, LVIHLQNCCP OVODHVO
PXY PUVHXCXH IVBDXIVZVGWP: (1) WQV PFPWVZPQNDSO AV, XC GNW WQVNIVWXHTSSF DGAIVTLTASV,
DGAIVTLTASV XG UITHWXHV; (2)HNZUINZXPV NC WQV PFPWVZ PQNDSO GNW XGHNGKVGXVGHV WQV
HNIIVPUNGOVGWP;(3) WQV LVF PQNDSO AV IVZVZAVITASV RXWQNDW GNWVP TGO PQNDSO AV
VTPXSFHQTGJVTASV; (4) WQV HIFUWNJITZP PQNDSO AV WITGPZXPPXASV AF WVSVJITUQ; (5)WQV TUUTITWDP NI
ONHDZVGWP PQNDSO AV UNIWTASV TGO NUVITASV AF T PXGJSVUVIPNG; (6) WQV PFPWVZ PQNDSO AV VTPF,
GVXWQVI IVBDXIXGJ LGNRSVOJV NC TSNGJ SXPW NC IDSVP GNI XGKNSKXGJ ZVGWTS PWITXG.WQVPV
IVBDXIVZVGWP PWXSS HNZUIXPV WQV XOVTS RQXHQ ZXSXWTIF HXUQVIP TXZTW. WQVF QTKV AVVG IVUQITPVO,
TGO BDTSXWXVP WQTW SXV XZUSXHXW QTKV -AVVGZTOV VYUSXHXW. ADW TGF ZNOVIG HIFUWNJITUQVI RNDSO
AV KVIF QTUUF XC TGFHXUQVI CDSCXSSVO TSS PXY.NC HNDIPV, XW QTP GVKVI AVVG UNPPXASV WN ON WQTW.
WQVIV TUUVTIP WN AV THVIWTXG XGHNZUTWXAXSXWF TZNGJ WQVZ WQTW ZTLVP XW XZUNPPXASV WN
XGPWXWDWVTSS NC WQVZ TW NGHV. WQV IVBDXIVZVGW WQTW XP DPDTSSF PTHIXCXHVO XP WQV
CXIPW.LVIHLQNCCP TIJDVO PWINGJSF TJTXGPW WQV GNWXNG NC T CXVSO HXUQVI WQTW RNDSOPXZUSF IVPXPW
PNSDWXNG SNGJ VGNDJQ CNI WQV NIOVIP XW WITGPZXWWVO WN AVHTIIXVO NDW. WQXP RTP GNW VGNDJQ,
QV PTXO, OVHSTIXGJ WQTW "WQV PVHIVWZTWWVI XG HNZZDGXHTWXNGP PVGW NKVI T OXPWTGHV KVIF NCWVG
IVWTXGP XWPXZUNIWTGHV AVFNGO WQV OTF NG RQXHQ XW RTP WITGPZXWWVO." QV RTP NG WQVPXOV NC WQV
TGJVSP, ADW T UITHWXHTS CXVSO HXUQVI WQTW XP DGAIVTLTASV RTPGNW UNPPXASV XG QXP OTF, GNI XP XW
WNOTF, TGO PN ZXSXWTIF HIFUWNJITUQF QTPPVWWSVO CNI CXVSO HXUQVIP WQTW OVSTF ADW ON GNW OVCVTW
HIFUWTGTSFPXP.UVIQTUP WQV ZNPW PWTIWSXGJ IVBDXIVZVGW, TW CXIPW JSTGHV, RTP WQV PVHNGO.
LVIHLQNCCP VYUSTXGVO WQTW AF "PFPWVZ" QV ZVTGW "WQVZTWVIXTS UTIW NC WQV PFPWVZ; WTASVTDY,
HNOV ANNLP, NI RQTWVKVIZVHQTGXHTS TUUTITWDP ZTF AV GVHVPPTIF," TGO GNW "WQV LVF UINUVI."LVIHLQNCCP
QVIV ZTLVP CNI WQV CXIPW WXZV WQV OXPWXGHWXNG, GNR ATPXH WNHIFUWNSNJF, AVWRVVG WQV JVGVITS
PFPWVZ TGO WQV PUVHXCXH LVF. RQF ZDPWWQV JVGVITS PFPWVZ "GNW IVBDXIV PVHIVHF," TP, CNI VYTZUSV,
T HNOVANNLIVBDXIVP XW? RQF ZDPW XW AV "T UINHVPP WQTW. . . NDI GVXJQANIP HTG VKVGHNUF TGO TONUW"?
AVHTDPV, LVIHLQNCCP PTXO, "XW XP GNW GVHVPPTIF WNHNGEDIV DU XZTJXGTIF UQTGWNZP TGO WN PDPUVHW
WQV XGHNIIDUWXAXSXWF NCVZUSNFVVP NI PDATSWVIGP WN DGOVIPWTGO WQTW, XC T PFPWVZ
IVBDXIXGJPVHIVHF RVIV XG WQV QTGOP NC WNN STIJV T GDZAVI NC XGOXKXODTSP, XW HNDSOAV HNZUINZXPVO
TW VTHQ VGJTJVZVGW XG RQXHQ NGV NI TGNWQVI NC WQVZWNNL UTIW." WQXP QTP UINKVO WN AV WIDV, TGO
LVIHLQNCCP' PVHNGOIVBDXIVZVGW QTP AVHNZV RXOVSF THHVUWVO DGOVI T CNIZ WQTW XP PNZVWXZVPHTSSVO
WQV CDGOTZVGWTS TPPDZUWXNG NC ZXSXWTIF HIFUWNJITUQF: WQTW WQVVGVZF LGNRP WQV JVGVITS PFPWVZ.
ADW QV ZDPW PWXSS AV DGTASV WN PNSKVZVPPTJVP XG XW RXWQNDW LGNRXGJ WQV PUVHXCXH LVF. XG
XWP ZNOVIGCNIZDSTWXNG, WQV LVIHLQNCCP ONHWIXGV PWTWVP WQTW PVHIVHF ZDPW IVPXOVPNSVSF XG
WQV LVFP.QTO LVIHLQNCCP ZVIVSF UDASXPQVO QXP UVIHVUWXNGP NC WQV UINASVZPCTHXGJ UNPW-WVSVJITUQ HIFUWNJITUQF
    TGO QXP UIVPHIXUWXNGP CNI IVPNSKXGJWQVZ, QV RNDSO QTKV TPPDIVO T USTHV CNI
QXZPVSC XG WQV UTGWQVNG NCHIFUWNSNJF. ADW QV OXO ZNIV. QV HNGWIXADWVO T WVHQGXBDV NC
HIFUWTGTSFPXPWQTW XP NC PDUIVZV XZUNIWTGHV WNOTF. HTSSVO "PDUVIXZUNPXWXNG,"
XWHNGPWXWDWVP WQV ZNPW JVGVITS PNSDWXNG CNI UNSFTSUQTAVWXH PDAPWXWDWXNGPFPWVZP. RXWQ
CVR VYHVUWXNGP, XW STFP GN IVPWIXHWXNGP NG WQV WFUV NI SVGJWQNC LVFP, TP ONVP WQV LTPXPLX
ZVWQNO, GNI NG WQV TSUQTAVWP, RQXHQ ZTF AVXGWVIIVSTWVO NI VGWXIVSF XGOVUVGOVGW. XW RTGWP NGSF
PVKVITS ZVPPTJVP XGWQV PTZV LVF. WQV HIFUWTGTSFPW ZDPW TSXJG WQVPV NGV TANKV WQV NWQVI PNWQTW
SVWWVIP VGHXUQVIVO RXWQ WQV PTZV LVFSVWWVI RXSS CTSS XGWN T PXGJSVHNSDZG. XG WQV PXZUSVPW
HTPV, WQTW NC T IDGGXGJ LVF (T KVIF SNGJHNGWXGDNDP WVYW DPVO TP T LVF, TP T GNKVS) WQTW IVPWTIWP
RXWQ VTHQZVPPTJV, QV HTG ON WQXP PXZUSF AF USTHXGJ TSS WQV CXIPW SVWWVIP XG WQV CXIPWHNSDZG,
TSS WQV PVHNGO SVWWVIP XG WQV GVYW HNSDZG, TGO PN NG.
```

Bibliography

[1] Technical University of Moldova. *Lucrare de laborator nr. 2: Criptanaliza cifrurilor monoalfabetice*. Course assignment, Sections 2.1–2.3 and variant V19, pp. 24–25.

[2] Crypto Corner. *Frequency Analysis: Breaking the Code*. Frequency-analysis and substitution activity specified by the assignment. Accessed October 14, 2026. `https://crypto.interactive-maths.com/frequency-analysis-breaking-the-code.html`.

[3] Pancenco, Mihail. *Variant 19 working notes*. Supplied ciphertext, frequency results and the recorded sequence of the first seven substitutions.